

Aircraft Notes

- Figure out spacing & amount of ribs by testing through simulation of design.

$$\frac{C_{L_{max}}}{C_{D_{max}}} = \frac{Z_{max}}{D_{max}} \quad \text{to figure out the lift needed to fly}$$

aircraft when given C_L & C_D from AoA. This are lift formula to find needed velocity

S-span

$$C_L = \frac{L}{q \cdot S}, \quad q = 0.5(\rho V^2 \cdot S)$$

$$L_{max} = W_{gross} \text{ during stabilization}$$

- Velocity needed to take off

$$V_{Takeoff} = \sqrt{\frac{2W}{\rho S C_L}}$$

- Approximation for takeoff

$$S_{Lo} = \frac{1.214 \cdot W^2}{g \cdot e \cdot S \cdot C_{L_{max}} \cdot T}$$